Day 1: Introduction to Network Protocols Theory

1. Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

Aim:-

To capture network traffic while accessing www.spotify.com using Wireshark and identify at least two DNS packets and two HTTP/HTTPS packets.

Theory:-

Wireshark is a network protocol analyzer used to capture and inspect packets transmitted over a network.

- DNS (Domain Name System): Converts a domain name, such as www.spotify.com, into an IP address.
- HTTPS (Hypertext Transfer Protocol Secure): Provides encrypted communication between the browser and the Spotify website.
- HTTP: Transfers web data without encryption. Spotify generally uses HTTPS for secure website communication.

Procedure:-

1. Open Wireshark on your computer.
2. Select your active Wi-Fi or Ethernet network interface.
3. Click Start Capturing Packets.
4. Open your browser and visit https://www.spotify.com.
5. Wait for the website to load, then return to Wireshark.
6. Enter dns in the display filter box and press Enter to view DNS packets.
7. Note down at least two DNS packets.
8. Change the filter to http or tls to identify HTTP or encrypted HTTPS-related traffic.
9. Record at least two relevant packets from your actual capture.

Observations:-

A. DNS Packets

Packet	Source	Destination	Protocol	Information
1	Your computer	DNS server	DNS	Query for www.spotify.com
2	DNS server	Your computer	DNS	DNS response containing an IP address or a CNAME record

B. HTTP/HTTPS Packets

Packet	Source	Destination	Protocol	Information
1	Your computer	Spotify server	TLS	Client Hello to initiate a secure connection
2	Spotify server	Your computer	TLS	Server response during the secure connection setup

2. Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

```
C:\Users\Prachi Patel>nslookup www.flipkart.com
Server:  UnKnown
Address:  192.168.0.1

Non-authoritative answer:
Name:    e348826.dsca.akamaiedge.net
Addresses:  49.44.140.178
          49.44.140.171
          49.44.140.169
          49.44.140.152
          49.44.140.177
          49.44.140.179
          49.44.140.147
          49.44.140.170
          49.44.140.160
Aliases:  www.flipkart.com
          www.flipkart.com.edgekey.net


C:\Users\Prachi Patel>
```

Output and Observation

Parameter	Observed result
Domain name	www.flipkart.com
DNS server	192.168.0.1
Resolved hostname	e348826.dsc.akamaiedge.net
IP address 1	49.44.140.178
IP address 2	49.44.140.171
IP address 3	49.44.140.169
IP address 4	49.44.140.152
Alias	www.flipkart.com

Explanation

The nslookup command queries a DNS server to find the IP address associated with www.flipkart.com. The returned IP address identifies a network endpoint that the browser can use to connect to the website. The address returned may vary depending on the DNS server and network configuration.

3. Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

HTTP stands for Hypertext Transfer Protocol, while HTTPS stands for Hypertext Transfer Protocol Secure. Both protocols are used for communication between a web browser and a web server.

Three Key Differences

Sr. No.	**HTTP**	**HTTPS**
1. Security	Data is transferred without encryption.	Data is encrypted using TLS to protect communication.
2. Port Number	Uses port 80 by default.	Uses port 443 by default.
3. Data Protection	Data can be easier to intercept and read during transmission.	Encryption helps prevent attackers from reading or modifying data in transit.

Real-World Example

Example: Flipkart

When I use the Flipkart website to shop online, it uses HTTPS to secure communication between my browser and its servers.

- It helps protect login credentials and personal information during transmission.
- It helps protect payment information from being read by someone intercepting the connection.
- The browser can verify the website's identity using its TLS certificate.

**4. Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.

Hint: Look for packets with 'Standard query' and 'Standard query response' in the Info column.**

Objective: To observe DNS (Domain Name System) queries and responses using Wireshark while accessing a new website.

Procedure:

1. Open Wireshark and select the active Wi-Fi or Ethernet network interface.
2. Start capturing network traffic.
3. Enter dns in the display filter bar and press Enter to display only DNS packets.
4. Open a browser and visit a website that you have not accessed recently, such as www.wikipedia.org.
5. Return to Wireshark and observe the packets in the Info column.
6. Look for packets labelled Standard query and Standard query response.
7. Stop the capture after the DNS packets appear.

Observations:

Packet	**Description**
Standard query	The computer sends a DNS request to find the IP address associated with the website's domain name.
Standard query response	The DNS server responds with the requested IP address or other DNS information.
A record	Provides the IPv4 address of the website.
AAAA record	Provides the IPv6 address of the website, if available.

Sequence of DNS Communication:

1. When I entered www.wikipedia.org in the browser, my computer generated a DNS query.
2. The DNS query was sent to a DNS server to resolve the domain name into an IP address.
3. Wireshark displayed the outgoing packet as Standard query in the Info column.
4. The DNS server returned a response containing the resolved IP address, which appeared as Standard query response.

5. After receiving the IP address, the browser used it to establish a connection with the website's server and load the webpage.

5. Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

Protocol Selected: DHCP (Dynamic Host Configuration Protocol)

Scenario: Connecting to Instagram using public Wi-Fi at a café

When I visit a café and connect my mobile phone to its Wi-Fi network to use Instagram, the DHCP protocol helps my phone connect to the internet.

Working Process:

1. Connection: I select the café's Wi-Fi network and enter the password.
2. DHCP Discover: My phone sends a request to find a DHCP server on the network.
3. DHCP Offer: The DHCP server offers an available IP address to my phone.
4. DHCP Request: My phone requests to use the offered IP address.
5. DHCP Acknowledgment: The DHCP server confirms the assignment and provides network settings, such as the subnet mask, default gateway, and DNS server.
6. Internet Access: My phone can now communicate over the network and access Instagram to view posts, stories, and reels.